Footprint White Paper: Versions 0.01–0.09

Oct 9, 2026

Summary

Footprint is a free, single-file browser app that turns live public trade data from Coinbase, Kraken and Bitstamp into an order-flow footprint chart. Version 0.09 is the current build: it covers six coins (BTC, ETH, SOL, XRP, LTC, DOGE), runs on desktop and Android, and needs no account, API key or server.

In plain terms: a normal candle tells you where price went. A footprint tells you who pushed it there. It splits each candle into price rows and shows how much was bought aggressively versus sold aggressively at each row.

Technically: the app opens public WebSocket trade streams, classifies each trade by aggressor side, and aggregates it into 1-minute base bars at a fixed per-coin price tick. Everything else is derived from those bars: display timeframes, row sizes, indicators and profiles. It renders on two HTML canvas layers and persists to IndexedDB (and optionally a local folder).

The paper has two layers throughout. Each version entry gives a plain-language description of what changed, then the technical detail. Nine versions are documented, 0.01 through 0.09, one per saved file.

Background: what a footprint chart shows

In plain terms. Every trade has a buyer and a seller, but one of them is in a hurry. The one in a hurry crosses the spread and takes the other side's resting order. That's the aggressor. If aggressive buyers outnumber aggressive sellers at a price, buying pressure won there. A footprint chart counts this at every price inside every candle, so you can see where the fight happened and who won.

Three things matter most when reading it:

- **Delta** is aggressive buys minus aggressive sells. Green rows mean buyers dominated; red rows mean sellers dominated.
- **Point of control (POC)** is the row with the most volume in a bar, outlined in gold.

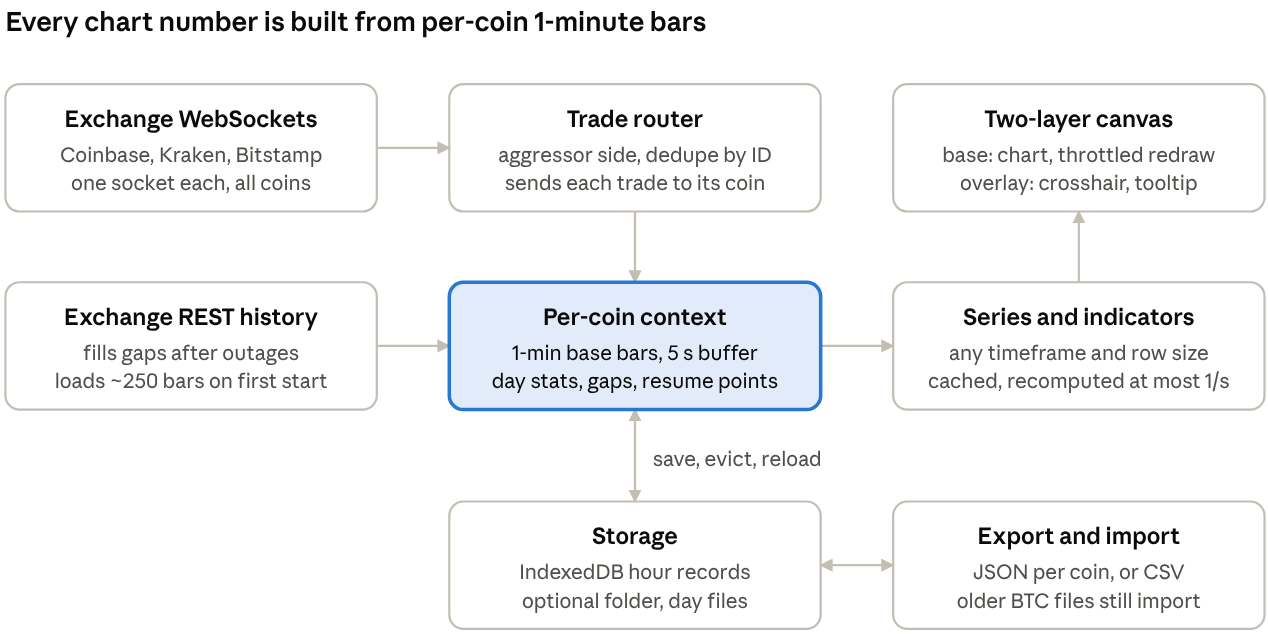
- **Absorption** is heavy aggression at a bar's high or low that failed to move price. It often shows a large passive player soaking up orders.

Technically. The app records each trade as (time, price, size, aggressor side). Coinbase reports the maker's side, so the app flips it. Kraken reports the taker's side directly, and Bitstamp's type field (0 = buy, 1 = sell) is treated as the taker side.

Trades are bucketed by time (the bar) and by price (the row). Each row stores two numbers: buy quantity and sell quantity. Delta, volume, POC, VWAP and every profile are sums over those cells. The chart never needs the raw trades again once they are aggregated.

Architecture (version 0.09)

All data flows one way, from exchange to per-coin 1-minute bars to the screen. Storage sits beside the bars, not in the path.



Footprint 0.09 architecture · data flow from exchange to screen

In plain terms. Trades come in, get sorted by coin and by who was aggressive, and are filed into one-minute buckets. Everything you see is computed from those buckets, so changing timeframe or row size never needs new data. Saving and reloading move whole hours of buckets in and out of the browser's database.

Technically. The bound context pattern lets the drawing code stay coin-agnostic: globals point at the viewed coin's context, and background coins are processed by briefly binding theirs. All network history jobs run through one serial queue to respect exchange rate limits (about 1.1 s between Kraken pages).

Version history

Footprint went from a basic live BTC footprint (0.01) to a six-coin, phone-ready app with storage and gap recovery (0.07), then phone fixes (0.08) and a layout, appearance and indicator update (0.09). Each version below matches a saved file.

Version	Headline change	Code size (lines)
0.01	First browser build: live footprint, profile, export/import	482
0.02	Restyle, ghost candles, first indicators, Auto Fib	723
0.03	Full indicator suite, bottom panels, settings, pause, speed work	1,115
0.04	Default profile (Auto Fib, SMA 200, custom-styled RSI)	1,138
0.05	Developer timeframes: 5, 10, 15, 30 seconds	1,169
0.06	Auto-save, gap backfill, single collecting tab, history on first start	1,694
0.07	Six coins, coin tracking, phone layout and touch, installable app	2,069
0.08	Phone fixes: screen kept on, Auto Fib off by default	2,075
0.09	Cleaner phone layout, day/night backgrounds, 11 new indicators, searchable menu	2,369

0.01: First browser build

Plain. Open the file in a browser and it starts listening to live BTC trades from Coinbase, Kraken and Bitstamp. Each candle is drawn as a stack of price rows showing buying versus selling, with a thin candle strip on the left edge of each column. A profile on the right

shows sell and buy volume per price for the visible range. Collected data can be exported and imported back later.

Technical. Three public WebSocket feeds, no keys: Coinbase Advanced Trade `market_trades`, Kraken WS v2 `trade`, and Bitstamp `live_trades_btcusd`. Trades aggregate into 1-minute base bars with $5 rows (`Map<rowIndex, [buy, sell]>`), then roll up to 1m–1h and $5–$500 rows. Cell modes are Delta, Sell × Buy and Volume, in USD or BTC; cell opacity scales with size against the 95th percentile. Same-side trades within 150 ms merge into one print for bubbles. A “Keep” setting (6–72 h) prunes old minutes. Export writes JSON (re-importable) or CSV, and the page warns before closing with unexported trades.

0.02: Restyle, candles and first indicators

Plain. The chart got a cleaner, more polished look. The bottom summary got bigger and easier to read, with more room between the chart and the profile. Faint candles now sit behind the cells, and hovering a bar shows its candle clearly. VWAP, EMAs and key levels were added, each with its own switch. Auto Fib draws retracements from the latest swing: yellow above price (resistance), purple below (support), flipping as price crosses them.

Technical.

- **Look:** tiered cell fills (neutral, tinted, solid) driven by a highlight threshold (auto = 88th percentile), a gold POC outline and a 13 px bar-delta strip. The profile became a delta profile with opacity ∝ √(row delta ÷ max).
- **Two canvases:** ghost candles render at 28% alpha on the base canvas, and the hovered candle on an overlay canvas, so hovering never forces a full redraw.
- **Indicators:** VWAP resets each UTC day from running sums of volume, price × volume and price² × volume, with ±σ bands. Also EMAs 9/21/50 and key levels (day open, day high/low, prior-day high/low).
- **Auto Fib:** anchored on the visible range's high and low; levels 0, 0.236, 0.382, 0.5, 0.618, 0.786, 1, colored against the last price on every draw.

0.03: Full indicator suite, settings, pause and speed

Plain. SMAs, Bollinger Bands and Ichimoku were added. A bottom panel area holds Volume and RSI (plus optional CVD, MACD and Stoch RSI), and panels resize by dragging. Every indicator has a settings window for lengths, colors and widths. Pause (or Space) freezes the chart while data keeps collecting. Hovering a profile row highlights every bar that traded there. The app was tuned to run lighter on older PCs.

Technical.

- **Indicators:** EMA 1–4 and SMA 1–3 slots, Bollinger (SMA ± kσ), Ichimoku with a cloud colored by span order, Wilder RSI, CVD (from visible start or each UTC day), MACD with a momentum-shaded histogram, and Stoch RSI.
- **Settings:** dialogs are generated from metadata tables (`IMETA`, `PMETA`). Auto Fib can anchor on the visible range or a fixed 4/12/24/48 h or custom lookback. Key levels can auto-flip between support and resistance colors.
- **Speed:** the base chart redraws only on data or view changes, throttled to 250–2,000 ms. The display series updates per trade instead of rescanning. Per-bar aggregates are cached, indicators recompute at most once a second, and drawing stops while the tab is hidden. A full redraw measured about 1.2 ms, an overlay redraw about 0.4 ms.

0.04: Default profile

Plain. One click loads a default setup: Auto Fib (yellow and purple), SMA 200 in white, and RSI with custom overbought and oversold colors, plus Volume. You can also save your own profile and load it back.

Technical. The RSI style is width 2, light blue in the normal range, red when overbought and green when oversold. It has dim grey dashed 70/30 lines, a fixed 0–100 scale, no signal arrows and the current value tagged on the axis. Other indicators default to a cool blue and grey palette clear of the support/resistance colors. Profiles live in `localStorage` (`fp_profile`).

0.05: Developer timeframes

Plain. 5-, 10-, 15- and 30-second charts were added, covering the last two hours, for quickly checking that everything works.

Technical. A second buffer holds 5-second bars for two hours, pruned every minute. Its cutoff is aligned to whole minutes so VWAP matches the 1-minute data exactly; tests confirmed identical VWAP across all nine timeframes.

0.06: Storage, gap recovery and history on first start

Plain. The app can save automatically, so RAM stays small and history survives restarts. It notices when it was closed, asleep or disconnected, fetches the missed trades from the exchanges' public history and marks every gap on the chart. Only one tab collects at a time. On first open, or when you change timeframe, it also loads about 250 bars of history instead of starting empty.

Technical.

- **Storage:** IndexedDB holds one record per UTC hour of 1-minute bars, saved every 60 s. Hours older than the RAM window (2–72 h) are evicted and reloaded when you scroll back. An optional folder mirror writes one JSON file per UTC day, and retention runs from 3 days to unlimited. Any hour held in memory is complete, so a save never overwrites stored data with a partial hour.
- **Gap recovery:** each venue keeps a resume point. On reconnect, a gap over 10 s triggers a backfill from Kraken REST (up to 7 days), Coinbase REST (24 h) or Bitstamp REST (24 h). Trade IDs remove duplicates. Gap markers show pending, filled, partial or missing.
- **Housekeeping:** a Web Lock makes one tab the collector. A late-firing 1 s timer detects sleep, and minimum trade lag flags a wrong PC clock.
- **History seeding:** 30 min for 5 s up to 240 h for 1 h. Each exchange fills what storage lacks within its reach (Kraken 10 days, others 24 h), with a chart marker where history becomes Kraken-only.

0.07: Six coins, phone version, installable app

Plain. A coin menu next to the “Footprint” title switches between BTC, ETH, SOL, XRP, LTC and DOGE. You choose whether only the viewed coin collects data or a selection keeps collecting in the background. On a phone the layout compacts and the chart responds to touch: pinch, swipe, hold for crosshair, tap for details, double-tap for live. A phone can be a light Viewer (the default) or a Collector. Hosted on a free site, the app installs to the home screen.

Technical.

- **Per-coin contexts:** each coin has its own bars, series, day stats, caches, gaps and resume points. Shared code runs against whichever context is bound.
- **Feeds:** one socket per exchange carries every tracked coin through subscribe and unsubscribe messages. Kraken DOGE tries `DOGE/USD`, then `XDG/USD`.
- **Per-coin price math:** base tick per coin (BTC $5, ETH $0.25, SOL and LTC $0.02, XRP $0.0002, DOGE $0.00002), eight row sizes and price decimals.
- **Storage v2:** records keyed by coin and hour. BTC data from 0.06 migrates automatically (verified in testing). Exports carry the coin, and old BTC-only files still import.
- **Phone:** pointer-event gestures, layout below 760 px, bottom-sheet menus, compact status line, battery saver (≤1 redraw/s, lower resolution) and an optional Wake Lock.
- **Installable app:** manifest, service worker (network-first for app files; never touches exchange data) and icons.

0.08: Phone fixes

Plain. Testing on a real phone found two problems. The screen went dark while viewing, and Auto Fib cluttered the small screen. Phones now keep the screen on by default, and Auto Fib starts off on phones (it's still one tap away under Indicators).

Technical. Wake Lock defaults on for phones via a one-time flag (`wakeSet`), so the user's later choice sticks. Because Android can drop the lock (power saving) or refuse it before the first touch, any tap re-requests it. The status line shows ☀ while it's held, or a note when the page isn't on https. "Load default" on a phone also leaves Auto Fib off. The service-worker cache version was bumped so installed copies update.

0.09: Cleaner phone layout, day/night backgrounds, more indicators

Plain. On the phone, the top bar now holds just the coin, the title, the timeframe and the menu. Pause moved to a floating button, with a Live button that appears only when you've scrolled back. Prices on the right are never cut off, even for coins with many decimals; the delta profile narrows instead. The background can follow the time of day, shifting from morning to evening and into a softer, easier-to-read night look. Eleven indicators were added, mostly order-flow tools, and the Indicators menu gained categories, search and favorites.

Technical.

- **Layout:** the price axis is sized each frame from the measured width of the coin's price labels (plus a 25% margin), and the profile column steps down from 92 px to 60 px with labels dropped before prices are squeezed. The status line collapses to one row on phones and expands on tap.
- **Themes:** every surface color comes from one token set (background top and bottom, panels, lines, grid, cells, text). Auto mode reads the device clock and interpolates tokens between four phases (default 6:00, 10:00, 17:00, 21:00) over 30 minutes. No location is used. Phones default to Auto; desktop keeps Midnight.
- **Order flow:** stacked imbalances compare buy volume at a row with sell volume one row below (and the reverse) at a set ratio (default 300%) over 3+ consecutive rows, with zones extended until price trades back into them. Value area expands from the profile's POC until 70% of volume is covered. Delta divergence flags a new N-bar high on negative delta (or low on positive delta).
- **Indicators:** anchored VWAP (from the lowest low or highest high in view, the day's start, or a bar picked on the chart), Supertrend (ATR 10 × 3), Keltner (EMA 20 ± 2× ATR 10), Donchian (20), classic pivots from the prior UTC day or week, prior-week high/low,

and OBV, ATR and ADX/DMI (Wilder smoothing) as bottom panels.

- **Menu:** 33 entries in six categories (Order flow, Volume, Trend, Volatility, Momentum, Levels), filtered by a search box that also matches descriptions, with starred favorites pinned to the top.
- **Performance:** a full redraw measured about 1.7 ms with the default setup and about 3–4 ms with every new indicator switched on.

Data sources and accuracy

Footprint shows spot trades from three US-accessible exchanges, not the whole market. Futures, perpetuals and offshore spot venues (Binance, Bybit, OKX) are not included, so totals are a sample of global flow.

Exchange	Live feed	Side reported	History reach (backfill / first load)	Status
Coinbase	Advanced Trade WebSocket, `market_trades`	Maker side, flipped to aggressor	24 h / 24 h	Side convention matches Coinbase docs
Kraken	WebSocket v2, `trade`	Taker side	7 days / 10 days	DOGE symbol tried as `DOGE/USD`, then `XDG/USD`
Bitstamp	WebSocket v2, `live_trades_<pair>`	`type` 0 = buy, 1 = sell (assumed taker)	Last 24 h only	Side mapping unverified on live data

In plain terms. Most numbers are as exact as the exchanges report them, with two caveats. First, if Bitstamp's buy/sell label means the opposite of what the app assumes, its delta would be inverted. The per-exchange delta in the status bar exists to catch that. Second, history older than 24 hours comes from Kraken alone, and the chart labels that boundary.

Technically.

- **Precision:** quantities are stored to 8 decimals. Row prices are exact multiples of each coin's base tick, so rows never drift.
- **Duplicates:** each venue keeps a set of the last 20,000–40,000 trade IDs, which removes repeats from reconnects and overlapping backfill pages.
- **Restarts:** a backfill interrupted by a shutdown is marked missing rather than retried, because retrying could double-count trades.
- **Untested live:** testing used stand-in exchange servers in a headless browser. Not yet exercised: live exchange feeds, cross-origin REST backfill from a `file://` page, the real folder picker, and the browser's persistent-storage grant.

Performance and resource use

A full chart redraw measured about 1.2 ms and an overlay redraw about 0.4 ms in testing (30 hours of simulated BTC trades, 1500 × 950 window). Memory is bounded by the RAM window you choose, not by how long the app runs.

In plain terms. The chart doesn't redraw for every trade. It redraws at most a few times a second (once a second in battery saver), while mouse and touch respond instantly. Older data moves to storage on its own schedule, so a session left running for days doesn't keep growing in memory.

Technically. These are the fixed limits that cap CPU and memory:

Limit	Value	Why
Redraw while streaming	250–2,000 ms (user setting); ⩾1,000 ms in battery saver	Caps CPU on old machines and phones
Canvas resolution	Device pixel ratio capped at 2 (1.5 in battery saver)	High-DPI phones otherwise draw up to 9× the pixels
RAM window per coin	2–72 h of 1-minute bars	Older hours are saved and evicted
5-second buffer	Last 2 h	Developer timeframes only
Large-print list	60,000 entries per coin	Bubbles stay bounded
Row-aggregate cache	6,000 bars	Cleared when exceeded

Limit	Value	Why
Duplicate-check IDs	20,000–40,000 per venue	Rolling window
Indicator recompute	≤1 per second for live updates	Long lookbacks such as SMA 200 stay cheap
Hidden tab	No drawing	Collection continues

Each coin tracked in the background adds its own socket subscription, memory and CPU use, roughly in proportion to that coin's trade rate.

Privacy and security

Footprint only reads public market data and never sends anything about you anywhere. It has no login, API key, wallet connection, analytics or server of its own.

In plain terms. The app listens to the same public trade tape anyone can see. It downloads public trade history when it needs to fill a gap. Your settings and chart history stay in your own browser (and, only if you choose a folder, in that folder). Hosting it on GitHub Pages, Netlify or Cloudflare publishes the app's code, not your data.

Technically.

- **Network, read-only:** three WebSocket endpoints and three public REST trade-history endpoints (Kraken, Coinbase, Bitstamp). No requests carry credentials or identifiers.
- **Local writes:** `localStorage` (settings, profile), IndexedDB (`btc-footprint`: hour records and meta), and — only after you pick a folder through the browser's picker — that one folder. The browser grants that folder's permission and can revoke it at any time.
- **No other computer access:** the page can't read other files or run programs.
- **Service worker (hosted copy only):** it caches only the app's own six files and never intercepts exchange traffic.

Known limitations and possible next steps

The biggest open item is confirming 0.09 against live markets on real devices. Everything else is a known trade-off.

- **Live verification:** Bitstamp side mapping, Kraken's DOGE symbol, REST backfill from a `file://` page, and the folder picker are untested against real services.
- **Coverage:** spot only, three US venues. No futures, perps or offshore books, and no order-book (resting liquidity) data.
- **History depth:** Coinbase and Bitstamp give 24 h of public history. Anything older is Kraken-only unless you collected it yourself.
- **Android background:** Chrome pauses background tabs. The app backfills on return, but a phone is not a reliable 24/7 collector.
- **Separate data per location:** the desktop file and the hosted copy keep separate histories; Export/Import moves data between them.
- **Absorption tag:** a heuristic (row delta ≥ 1.5× threshold at the bar's outer 30%, with price closing at least 2–3 rows away), not a confirmed signal.

Possible next steps, none committed:

- [ ] Live test on desktop and phone
- [ ] Confirm Bitstamp side mapping from the per-venue delta
- [ ] Consider adding a perpetuals feed for comparison

Glossary

Term	Meaning
Aggressor	The side of a trade that crossed the spread (market order)
Delta	Aggressive buy volume minus aggressive sell volume
Footprint	A candle split into price rows, each showing buy and sell volume
POC	Point of control: the price row with the most volume
Delta profile	Delta and volume summed per price row across a range of bars
Absorption	Heavy aggression at a bar extreme that price failed to follow

Term	Meaning
CVD	Cumulative volume delta: running total of delta
VWAP	Volume-weighted average price, reset each UTC day
Base tick	The finest stored price row for a coin (BTC $5)
Base bar	The stored 1-minute unit every timeframe is built from
Backfill	Fetching missed trades from an exchange's public history
Seed	The history loaded on first start for the chosen timeframe
Collector / Viewer	Device roles: saves history / streams and keeps nothing
IndexedDB	The browser's built-in database where history is saved
Service worker	Browser script that lets the hosted app open offline and install
Wake Lock	Browser feature that keeps the screen on
Stacked imbalance	Several consecutive price rows where one side's aggression outweighs the other by a set ratio (default 300%)
Value area (VAH / VAL)	The price range holding 70% of volume around the POC; VAH and VAL are its top and bottom
Anchored VWAP	VWAP calculated from a chosen starting bar instead of the day's start
ATR	Average true range: typical bar size, a measure of volatility
ADX / DMI	Trend strength (ADX) and direction (+DI vs −DI)